An explicit dual window for Hermite Gabor superframes

Candidate full answer to the non-explicitness remark in arXiv:1012.4283

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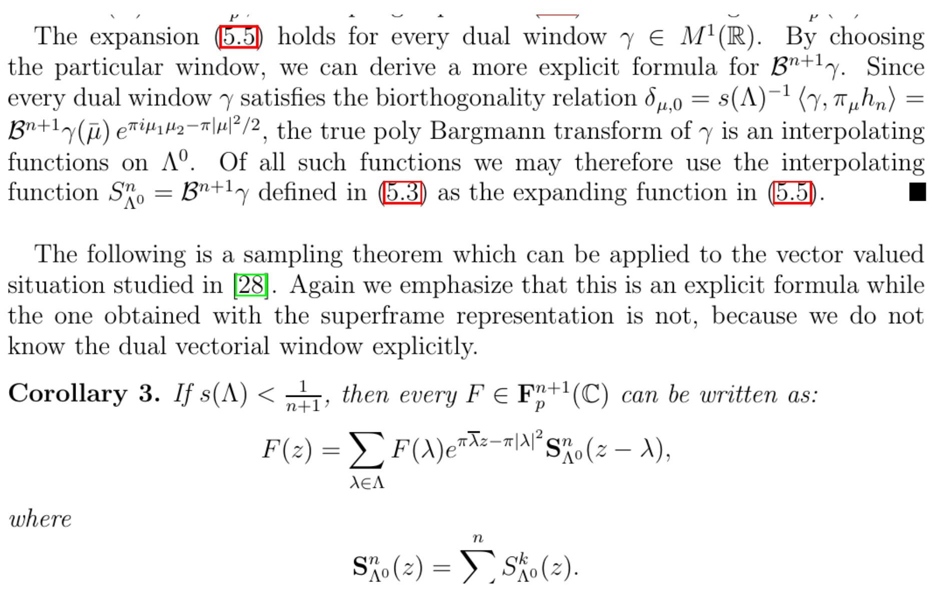
Result. Let $\mathbf{h} = (H_0, \dots, H_n)$ be the vector of the first $n+1$ normalized Hermite functions, and let Λ be a lattice with $s(\Lambda) < 1/(n+1)$. A dual vector window $\boldsymbol{\gamma} = (\gamma_0, \dots, \gamma_n)$ is given explicitly below. Its ordinary Bargmann transforms are a modified Weierstrass sigma function times finite Taylor polynomials. Every component belongs to the Feichtinger algebra M^1, the full matrix Wexler–Raz identities hold, and the resulting superframe reconstruction transfers to the vector modulation spaces and to an explicit polyanalytic sampling kernel.

1 The source remark

Abreu asks for an explicit dual vectorial window after deriving a sampling formula by decomposing the full polyanalytic Fock space into its true polyanalytic summands [1]. The earlier Hermite-superframe paper reduces the dual-window problem to a finite Hermite interpolation problem and gives a triangular ansatz, but leaves its coefficients implicit [2]. The formula below solves that triangular system in closed form.



The expansion (5.5) holds for every dual window $\gamma \in M^1(\mathbb{R})$. By choosing the particular window, we can derive a more explicit formula for $\mathcal{B}^{n+1}\gamma$. Since every dual window γ satisfies the biorthogonality relation $\delta_{\mu,0} = s(\Lambda)^{-1}\langle \gamma, \pi_\mu h_n \rangle = \mathcal{B}^{n+1}\gamma(\bar{\mu})\, e^{\pi i \mu_1 \mu_2 - \pi|\mu|^2/2}$, the true poly Bargmann transform of γ is an interpolating functions on Λ^0. Of all such functions we may therefore use the interpolating function $S^n_{\Lambda^0} = \mathcal{B}^{n+1}\gamma$ defined in (5.3) as the expanding function in (5.5). ■

The following is a sampling theorem which can be applied to the vector valued situation studied in [28]. Again we emphasize that this is an explicit formula while the one obtained with the superframe representation is not, because we do not know the dual vectorial window explicitly.

Corollary 3. *If $s(\Lambda) < \frac{1}{n+1}$, then every $F \in \mathbf{F}^{n+1}_p(\mathbb{C})$ can be written as:*

$$F(z) = \sum_{\lambda \in \Lambda} F(\lambda) e^{\pi \overline{\lambda} z - \pi|\lambda|^2} \mathbf{S}^n_{\Lambda^0}(z - \lambda),$$

where

$$\mathbf{S}^n_{\Lambda^0}(z) = \sum^{n} S^k_{\Lambda^0}(z).$$

Figure 1: The explicitness remark in arXiv:1012.4283, page 15.

2 Notation and formula

Put $N = n+1$. Let Λ° be the adjoint lattice and $\Omega = \overline{\Lambda^\circ}$. Let σ_Ω be the modified Weierstrass sigma function used in [2]; it has simple zeros precisely at Ω and satisfies the standard lattice growth estimate. Define

$$S(z) = \sigma_\Omega(z)^N, \qquad q(z) = \frac{S(z)}{z^N},$$

where the singularity of q at zero is removed. Thus $q(0) \neq 0$, and q has a zero of order N at every point of $\Omega \setminus \{0\}$.

For a function a holomorphic near zero, write

$$[a]_r(z) = \sum_{k=0}^{r} \frac{a^{(k)}(0)}{k!} z^k$$

for its degree-r Taylor truncation. For $0 \le j < N$, set

$$E_j(z) = \frac{z^j}{j!}\, q(z)\, [q^{-1}]_{N-1-j}(z), \tag{1}$$

and

$$G_j(z) = s(\Lambda)\sqrt{\pi^j j!}\, E_j(z), \qquad \gamma_j = \mathcal{B}^{-1} G_j. \tag{2}$$

Here $\mathcal{B}$ is the ordinary unitary Bargmann transform, with the conventions of [2]. Equations (1)–(2) are the promised explicit dual window.

For example, when $N = 2$ and $q(z) = q_0 + q_1 z + O(z^2)$,

$$G_0 = s(\Lambda) q(z) \left(\frac{1}{q_0} - \frac{q_1}{q_0^2} z \right), \qquad G_1 = s(\Lambda)\sqrt{\pi}\, \frac{z q(z)}{q_0}.$$

3 Proof

Lemma 1 (Cardinal jets)**.** *For $0 \le j, \ell < N$,*

$$\begin{aligned} &E_j^{(\ell)}(0) = \delta_{j\ell}, \\ &E_j^{(\ell)}(\omega) = 0 \qquad\qquad\qquad (\omega \in \Omega \setminus \{0\}). \end{aligned}$$

Proof. Taylor multiplication gives

$$q(z)[q^{-1}]_{N-1-j}(z) = 1 + O(z^{N-j}).$$

Consequently $E_j(z) = z^j/j! + O(z^N)$, which is exactly the first assertion. At a nonzero $\omega \in \Omega$, the function q has a zero of order N; the remaining factors in (1) are holomorphic there. Hence E_j also has a zero of order at least N, proving the second assertion. □

Lemma 2 (Feichtinger regularity)**.** *Each γ_j belongs to $M^1(\mathbb{R})$.*

Proof. The modified sigma estimate from [2] is

$$|S(z)| \lesssim \exp\left(\frac{\pi N s(\Lambda)}{2} |z|^2 \right).$$

Outside a fixed disk, (1) is a finite sum of terms $c_k S(z) z^{j+k-N}$ with $j + k \le N - 1$; thus it has at most a harmless polynomial factor times the displayed Gaussian. On the disk it is entire and bounded. Since $Ns(\Lambda) < 1$,

$$\int_{\mathbb{C}} |G_j(z)| e^{-\pi|z|^2/2} \, dm(z) < \infty.$$

The Bargmann characterization of the Feichtinger algebra therefore gives $\gamma_j \in M^1(\mathbb{R})$. □

Theorem 1 (Explicit Hermite superframe dual)**.** *The vector* $\boldsymbol{\gamma} = (\gamma_0, \ldots, \gamma_{N-1})$ *defined by* (1)–(2) *is a dual window for* $\mathcal{G}(\mathbf{h}, \Lambda)$ *in* $L^2(\mathbb{R}, \mathbb{C}^N)$*.*

Proof. For $\mu \in \Lambda^\circ$ and $0 \le j, \ell < N$, the Bargmann–Hermite identity in [2] reads

$$\begin{aligned} \langle \gamma_j, \pi_\mu H_\ell \rangle = {} & e^{-i\pi\mu_1\mu_2} e^{-\pi|\mu|^2/2} \frac{1}{\sqrt{\pi^\ell \ell!}} \\ & \times \sum_{k=0}^{\ell} \binom{\ell}{k} (-\pi\mu)^k G_j^{(\ell-k)}(\overline{\mu}). \end{aligned} \tag{3}$$

If $\mu \ne 0$, then $\overline{\mu} \in \Omega \setminus \{0\}$, and every derivative appearing in (3) vanishes by Lemma 1. If $\mu = 0$, only the $k = 0$ term remains and Lemma 1 gives

$$\langle \gamma_j, H_\ell \rangle = \frac{G_j^{(\ell)}(0)}{\sqrt{\pi^\ell \ell!}} = s(\Lambda)\delta_{j\ell}.$$

Therefore

$$\frac{1}{s(\Lambda)} \langle \gamma_j, \pi_\mu H_\ell \rangle = \delta_{\mu 0} \delta_{j\ell},$$

which is the full matrix Wexler–Raz condition. Lemma 2 ensures that all component Gabor systems are Bessel. The vector Wexler–Raz theorem now proves that $\boldsymbol{\gamma}$ is a dual window. □

4 Explicit superframe sampling kernel

Let $\mathcal{B}^{j+1}$ denote the true polyanalytic Bargmann transform associated with H_j. Write

$$\mathbf{B}^N(f_0, \ldots, f_{N-1}) = \sum_{j=0}^{N-1} \mathcal{B}^{j+1} f_j.$$

Define

$$\Psi_\Lambda^N(z) = \sum_{j=0}^{N-1} \mathcal{B}^{j+1}\gamma_j(z) = \sum_{j=0}^{N-1} \left(\frac{\pi^j}{j!}\right)^{1/2} e^{\pi|z|^2} \left(\frac{d}{dz}\right)^j \left[e^{-\pi|z|^2} G_j(z)\right]. \tag{4}$$

This is a finite expression in the modified sigma function, its reciprocal Taylor coefficients at zero, and derivatives.

Corollary 1 (Superframe reconstruction)**.** *If* $F = \mathbf{B}^N \mathbf{f}$*, then*

$$F(z) = \sum_{\lambda \in \Lambda} F(\lambda) e^{\pi\overline{\lambda}z - \pi|\lambda|^2} \Psi_\Lambda^N(z - \lambda). \tag{5}$$

For $1 \le p < \infty$*, the corresponding vector Gabor expansion converges in the coordinatewise vector modulation space* $M^p(\mathbb{R}, \mathbb{C}^N)$*; for* $p = \infty$ *it converges weak-*.*

Proof. Theorem 1 gives

$$\mathbf{f} = \sum_{\lambda \in \Lambda} \langle \mathbf{f}, \pi_\lambda \mathbf{h} \rangle \, \pi_\lambda \boldsymbol{\gamma}.$$

The polyanalytic Bargmann transform converts the coefficient into the weighted sample of F and intertwines time–frequency shifts with Bargmann–Fock shifts. Applying $\mathbf{B}^N$ yields (5) and (4). Since all H_j and γ_j belong to M^1 and there are only finitely many components, the usual bounded analysis and synthesis maps on modulation spaces give norm convergence for finite p and weak-* convergence at the endpoint. □

5 What is new and what is inherited

The density threshold, sigma-function growth estimate, Bargmann–Hermite identity, vector Wexler–Raz theorem, and existence of a solution via a triangular linear system all come from [2]. The new step is the closed formula (1): multiplication by a finite Taylor truncation of $1/q$ solves all N^2 jet conditions simultaneously. It converts the implicit triangular construction into an explicit vector window and hence supplies the superframe reconstruction that arXiv:1012.4283 called non-explicit. This is not a formula for the canonical dual; it is one explicit dual with Gaussian-type phase-space decay.

References

[1] L. D. Abreu, *Banach Gabor frames with Hermite functions: polyanalytic spaces from the Heisenberg group*, arXiv:1012.4283.

[2] K. Gröchenig and Y. Lyubarskii, *Gabor (Super)Frames with Hermite Functions*, Math. Ann. 345 (2009), 267–286; arXiv:0804.4613.